

MANRAJ SINGH

Permanent Resident - **Canada**

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PROFILE

Senior Software Engineer with 7.2 years building backend and distributed systems, including 3.5 years at Amazon. Recent work spans cloud application architecture on AWS, real-time protocol design over WebSockets and gRPC, and retrieval-augmented generation over large numbers of documents. Comfortable owning a service end to end, from API design through deployment, and cutting latency on systems already in production.

CORE COMPETENCIES

- Languages: Java (Advanced), TypeScript, Python, Node.js, Kotlin, C++
- Backend: Spring Boot, Node.js, gRPC, WebSockets, REST APIs, Microservices
- Frontend: React, Angular, Redux, TypeScript, Electron, HTML/CSS
- Databases: PostgreSQL, MySQL, MongoDB, DynamoDB, OpenSearch
- Cloud (AWS): EKS, ECS, Lambda, EC2, S3, EFS, ECR, RDS, DynamoDB, API Gateway, CloudFront, Bedrock, WAF, CloudFormation, CDK
- DevOps & CI/CD: Docker, GitHub Actions, Cypress, Maven
- AI/ML: Retrieval-Augmented Generation, Vector Search, Embeddings, Amazon Bedrock, MCP servers.
- Architectures: Distributed Systems, Microservices, Cloud-Native, Event-Driven
- Performance: Latency Reduction, High-Throughput API Design

WORK EXPERIENCE

Senior Software Engineer

Ansys (Acquired by Synopsys), Canada

September 2025 – Current

- **AEDT Doc Copilot - Retrieval-Augmented Generation (RAG) Assistant for Product Documentation**
 - Built a RAG pipeline over the Ansys documentation corpus so customers can ask questions in natural language from a chat panel inside the application, instead of searching the docs or filing a support ticket.
 - Designed the ingestion pipeline: documents are parsed, chunked along section boundaries, embedded, and indexed for hybrid retrieval, so a query matches on meaning and on exact keyword together.
 - Grounded every answer in retrieved passages with citations linking back to the source page, and had the assistant decline when retrieval returns nothing relevant rather than guessing.
 - Deployed on AWS behind CloudFront and API Gateway, with the vector index in OpenSearch Serverless and generation through Amazon Bedrock. Infrastructure defined in CDK
 - Secured the public endpoint with SSO token verification and per-tenant entitlement checks, with AWS WAF rate limiting to bound query volume per user and cap cost exposure on a metered LLM path.
 - **Tech Stack:** Python, TypeScript, Amazon Bedrock, RAG, Vector Search, Embeddings, OpenSearch, AWS (Lambda, API Gateway, CloudFront, S3, WAF), CDK, GitHub Actions
- **Concept SI - Web Platform for 3D Circuit Board Simulation**
 - Moved a Windows-only desktop tool to the web. The simulation engines already existed as native services. I built a layer that lets a browser drive them.
 - Designed a WebSocket protocol carrying approx 30 operations, translated to gRPC on the backend. Requests are matched by ID so many can run at once over one connection.
 - Design files run past 500 MB, so services never send them to each other. Files land once on a shared EFS volume and every service reads from the same path.
 - Gave each user their own copy of a file, keyed by their SSO identity. Two people can open the same design without conflicting, so no locking was needed.
 - Secured every backend call with SSO token verification, plus guards against SSRF, path traversal, and NoSQL injection.

- Packaged every service as a container image and deployed the stack to a managed cluster on AWS, with per-environment configuration so cloud and on-prem installs run from the same images. Builds publish through GitHub Actions on merge.
- Packaged the same stack as an Electron desktop app so customers who can't put data in the cloud run everything locally. Only the configuration changes.
- **Tech Stack:** Node.js, TypeScript, gRPC, WebSockets, Angular, MongoDB, Docker, AWS (EKS, ECR, EFS, S3, CloudFront), GitHub Actions

Software Development Engineer

Amazon, Canada

May 2022 – September 2025

- **BranchyAI – AI-Powered Ticket Resolution System**

- Reduced AI hallucinations in internal ticket resolution by building a decision-tree UI with **Next.js** and **React Flow**, mapping team → classification → resources.
- Integrated with **MCP servers** to auto-resolve tickets by connecting AI agents to internal tools based on classification indicators which are small prompts for each classification.
- Designed REST APIs to retrieve team-specific decision trees and resolution resources, enabling AI to dynamically fetch context.
- Improved resolution accuracy and automation coverage for redundant tickets in internal Amazon support workflows.

- **MultiCreate - Bulk Listing Creation with a Hybrid Draft Store**

- Built a flow where sellers create up to 20 products in one window, picking a base product to clone from a generative AI search that matches on uploaded images or a text description.
- Images upload straight from the browser to S3 over presigned URLs, in parallel, so image data never passes through the service.
- Drafts range from 10 KB to a few MB and are stored server-side, split by size: under **380 KB** compressed inline in DynamoDB, above that in S3 behind a pointer. About 95% stay inline, so most loads finish in one hop.
- Every draft carries a version, checked on write, so two windows editing the same product can't silently overwrite each other.
- Saves run in the background on the product switch, so moving between products never waits on a save.
- Drafts persist for 7 days from the last edit and reopen on any browser or machine the seller signs in on.
- Handles **~6000 TPS**, with draft loads around **50ms** from DynamoDB and **250ms** from S3.
- Tech Stack: Java, Spring Boot, DynamoDB, S3, Presigned URLs, Generative AI, REST APIs, React, TypeScript, CloudWatch

- **Keyword Search Redesign - API Split, Infinite Scroll, Legacy Service Deprecation**

- Inherited keyword search from another team at **~5s P99**. Traced it to an upstream aggregator resolving per-seller eligibility for every result before returning anything, so page size was effectively the latency dial.
- Split the read path into a lightweight discovery call and an on-demand detail call. Eligibility drops from 20 calls per search to 1 on selection, decoupling first render from page size. **P99 ~600-700ms** on search, **~300-500ms** on detail.
- The action link ships in the same payload as the restriction messages, so a listing cannot be cloned without its eligibility having been fetched. The gate is enforced at the API boundary, not in UI logic.
- Replaced fixed 20-result pages with infinite scroll via a custom React hook on an IntersectionObserver sentinel, cursor-paged 10 at a time up to 1,000 results.
- Cancelled in-flight requests with AbortController. Infinite scroll appends rather than replaces, so a stale response from an abandoned search corrupts the list instead of harmlessly overwriting it.
- Migrated sellers off the legacy endpoint behind a weblab, confirmed zero remaining consumers on usage dashboards, and deprecated the aggregator.
- Handles **~7,500 TPS** at peak.
- Tech Stack: Java, Spring Boot, REST APIs, React, TypeScript, IntersectionObserver, AbortController, CloudWatch

- **Global Search - Cross-Region Catalog Lookup API:**
 - Built an API that looks up 20 product IDs at once. It checks two catalogs in parallel, the caller's own region and a US fallback, instead of checking one and then the other. Most lookups miss in the home region, so this cuts out a second round trip on most requests. Handles **~9000 TPS**.
 - The fallback created a compliance gap. A product found in the US may be illegal to sell in the caller's country. No single service could catch this, because only the layer that merges the results knows where a result came from and who asked for it. I added the check there. One bulk call covers all 20 products instead of one call each.
 - Cached the compliance result by product and marketplace with a short TTL. Legality does not change per seller, so I left the seller out of the key. One cached entry then serves every caller in that region.
 - If the compliance check times out, the API returns nothing and says why. A failed search can be retried. An illegal listing cannot be undone that easily.
 - Per-seller permission checks happen later, when a user clicks a result. That keeps the list path free of per-item downstream calls.
 - **Tech Stack:** Java, Spring Boot, REST microservices, React, TypeScript
- **POD-Based Routing & Cross-Region API Architecture:** Migrated from merchant-based to a seller-centric POD model, improving routing consistency and reducing regional dependency. Built multi-region clients for NA, EU, and FE, achieving 1-second average latency and supporting **5000 TPS** post-migration.
 - **Tech Stack:** Java, Spring Boot, REST APIs, Distributed Microservices, AWS (multi-region clients for NA, EU, FE)
- **Generative AI for Product Description Automation (Amazon SellerCentral):** Developed a Generative AI solution using Amazon Bedrock to automatically generate product descriptions, reducing listing creation turnaround time by **~50%**. Designed backend APIs and a frontend UI for seamless integration with the seller experience.
 - **Tech Stack:** Amazon Bedrock, Generative AI, React, TypeScript, Java, Spring Boot, TanStack Router, REST APIs
- **CI/CD Automation:** Architected and implemented a robust CI/CD pipeline with multiple deployment stages (Beta, Gamma, Prod). Integrated Cypress tests to automate end-to-end testing.
 - **Tech Stack:** Java, CDK, Cypress, TypeScript, JavaScript
- **Mentorship & Collaboration Initiative:** Mentored two new hires and one intern, leading the migration of legacy Java Spring Boot applications. Improved migration time through process improvements and enhanced team collaboration.
 - **Tech Stack:** Java, Spring Boot
- **JDK-17 Migration:** Migrated from JDK 8 and JDK 11 to JDK 17, reducing garbage collection (GC) time by **84%**, CPU usage by **50%**, and memory consumption by **30%**, leading to significant application performance improvements.
 - **Tech Stack:** Java, JDK 17

Lead Teaching Assistant

Dalhousie University, Halifax, Canada

Sep 2021 – Apr 2022

- Led 4 TAs and 5 markers, supervised 24 student project groups, and built a Java tool that automated assignment distribution

Software Engineer

Amdocs, Pune, India

Jul 2019 – Apr 2021

- **API Development:** Engineered Spring Boot APIs for order orchestration, achieving a 95% accuracy in delivered story points across sprints.
- **System Optimization:** Enhanced a cloud capacity report microservice with algorithmic improvements, resulting in a 126% efficiency gain over the previous version.
 - **Tech Stack:** Java, Spring Boot, Big Data, Networking, Linux
- **Debugging and RCA:** Conducted comprehensive root cause analyses, resolving issues and providing optimized solutions with up to 30% faster processing times.

- **Stakeholder Engagement:** Delivered demos of developed solutions to stakeholders, fostering better communication and aligning deliverables with business objectives using written communication and collaboration skills.

Graduate Engineer Trainee

Synopsys, Noida, India

Jun 2019 – Jul 2019

- Automation Scripting: Developed a script to automate manual chip testing tasks, reducing effort by 50% and improving report generation accuracy. **Tech Stack:** Python, Linux, System Administration, C++
- Product Support: Worked on the VC Spyglass CDC Synchronizer module, testing chip circuit designs to ensure adherence to quality standards. **Tech Stack:** C, C++, Python

PROJECTS

- **Designing and implementing RDBMS in JAVA from scratch:** [Link](#)
 - **Tech Stack:** Java, Design Patterns, Data Structures, TDD, Maven
- **Cab booking Management:** [Link](#)
 - **Tech Stack:** Java, SOLID Principles, Design Patterns, Test Driven Development, Maven, CI/CD, GitFlow, AWS SNS
- **Hiree: (Online job posting and finding portal)** [Link](#)
 - **Tech Stack:** Java, Spring Boot, Docker, Maven, AWS (EC2, SNS, RDS, S3).
- **Volunteer Mart:** [Link](#)
 - **Tech Stack:** React, Node.js, GCP (Relational Database)

EDUCATION

Master of Applied Computer Science

May 2021 – Aug 2022

Dalhousie University, Halifax, Canada

- Acted as **Lab Assistant and Project Coordinator** for the **Data Management and Warehousing** course, delivering labs and managing project groups for **2 consecutive terms**

Bachelor of Engineering

Aug 2015 – Jun 2019

Thapar Institute of Engineering and Technology (TIET), Punjab, India